



C++ ROADMAP

Step by Step to Become a C++ Developer



Learn. Practice. Build. Master C++.

Strong foundation today, limitless possibilities tomorrow.

```
#include <iostream>
using namespace std;
int main() {
    cout << "Hello C++!";
    return 0;
}
```



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1. Getting Started

- What is C++?
- Features and Applications
- Setup IDE (VS Code / Code::Blocks / Visual Studio)
- Your First C++ Program

⌚ 1-2 Days

OUTCOME

Write and run your first C++ program.



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2. Basics of C++

- Data Types, Variables, Constants
- Input / Output (cin, cout)
- Operators and Expressions
- Type Casting

⌚ 2-3 Days

OUTCOME

Understand the basic building blocks of C++.



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3. Control Flow

- if, else, else if
- Switch Case
- For, While, Do-While Loops
- Break, Continue, Goto

⌚ 3-4 Days

OUTCOME

Control the flow of your programs.



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4. Arrays & Strings

- One-dimensional / Multi-dimensional Arrays
- Strings and String Functions
- Character Handling

⌚ 3-4 Days

OUTCOME

Work with collections of data efficiently.



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5. Functions

- Define and Call Functions
- Function Overloading
- Default Arguments
- Recursion

⌚ 4-5 Days

OUTCOME

Write modular and reusable code.



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6. OOPs Concepts

- Classes and Objects
- Constructors and Destructors
- Access Specifiers (public, private, protected)
- this Pointer

⌚ 1-2 Weeks

OUTCOME

Build programs using Object-Oriented principles.



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7. Advanced OOPs

- Inheritance (Single, Multiple, Multilevel, Hierarchical)
- Polymorphism (Function Overloading, Overriding)
- Abstraction (Abstract Classes)
- Encapsulation

⌚ 1-2 Weeks

OUTCOME

Write efficient and reusable object-oriented programs.



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8. STL (Standard Template Library)

- Containers (vector, list, set, map, stack, queue)
- Iterators
- Algorithms (sort, search, etc.)

⌚ 1 Week

OUTCOME

Use STL to solve problems faster and smarter.



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9. File Handling & Exceptions

- File I/O (fstream)
- Exception Handling (try, catch, throw)
- Custom Exceptions

⌚ 3-4 Days

OUTCOME

Handle files and errors like a pro.



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10. Practice & Build Projects

- Solve coding problems (DSA with C++)
- Build Mini Projects
- Work on Real-world Projects

⌚ Ongoing

OUTCOME

Build strong logic and real-world experience.



RECOMMENDED TOOLS



VS Code



Code::Blocks



CLion



LeetCode

PROJECT IDEAS

- ✓ Calculator
- ✓ Student Management System
- ✓ Library Management System
- ✓ Banking System
- ✓ Tic Tac Toe / Games



TIPS FOR SUCCESS

- ✓ Practice every day
- ✓ Solve problems on LeetCode / Codeforces
- ✓ Read and understand code
- ✓ Build projects
- ✓ Be consistent and patient



Consistency is the key! Keep coding, keep learning, and enjoy the journey.

